PROBLEM 15

Stokes Drag and Terminal Velocity

Original statement · Problems.pdf, p. 9

Viscous Aerodynamics

The Stokes' law states that the magnitude of the drag force acting on a sphere of radius r and velocity v moving on a fluid at rest of dynamic viscosity μ is: $F(v) = 6\pi\mu r v$. Assuming the density of a vertically-falling sphere is twice the density of the fluid (ρ_0), and considering drag, the weight, and the buoyancy of the sphere, compute:

(a) The terminal velocity of the sphere.

(b) The evolution of the position as a function of time, if the sphere starts from rest.

Remember that the kinematic viscosity is defined as $\nu = \mu/\rho_0$.

Worked Solution

Constraints, Degrees of Freedom & Coordinates

- **System:** a sphere of radius r and density $\rho_s = 2\rho_0$ falls vertically from rest in a fluid at rest of density ρ_0 and dynamic viscosity μ. The forces are the weight, the buoyancy and the Stokes drag $F_D = 6\pi\mu r v$ (as the statement says; unsteady and added-mass effects are not considered).
- **Constraints:** the motion is vertical (the sphere does not rotate or drift sideways); $\mathrm{CDOF} = 1$, with the coordinate x measured downward from the release point.
- **Initial conditions:** $x(0) = 0, \dot{x}(0) = 0$.

- **Relations:** volume $V = \frac{4}{3}\pi r^3$; mass $m = \rho_s V = \frac{8}{3}\pi\rho_0 r^3$; kinematic viscosity $\nu = \mu/\rho_0$, so $\mu = \rho_0\nu$.

Reference Frames & Vector Bases

- **Inertial frame:** $\mathcal{F}_0$ fixed to the fluid, with Ox pointing downward along gravity ($\mathbf{i}$ downward); constant unit vectors.

Derivation

3.1 Kinematics

$$\mathbf{r} = x\,\mathbf{i}, \qquad \mathbf{v} = \dot{x}\,\mathbf{i}, \qquad \mathbf{a} = \ddot{x}\,\mathbf{i}$$

3.2 Forces

Pedagogical Justification: three forces act along Ox (positive downward). The drag opposes the motion, which is downward ($\dot{x} > 0$), so it points upward.

$$W = mg = \tfrac{8}{3}\pi\rho_0 r^3 g\ (\downarrow), \qquad F_B = \rho_0 V g = \tfrac{4}{3}\pi\rho_0 r^3 g = \tfrac{1}{2}W\ (\uparrow), \qquad F_D = 6\pi\mu r\dot{x} = 6\pi\rho_0\nu r\dot{x}\ (\uparrow)$$

- **Net weight corrected by buoyancy:** $W - F_B = \frac{4}{3}\pi\rho_0 r^3 g = \frac{1}{2}mg$. **Inertia forces:** none.

3.3 Equations of Motion

- **Newton's second law along Ox:** $m\ddot{x} = W - F_B - F_D$:

$$\tfrac{8}{3}\pi\rho_0 r^3\,\ddot{x} = \tfrac{4}{3}\pi\rho_0 r^3 g - 6\pi\rho_0\nu r\,\dot{x}$$

Dividing by $\frac{8}{3}\pi\rho_0 r^3$: the first term gives $\dfrac{4/3}{8/3}g = \dfrac{g}{2}$ and the second $\dfrac{6\pi\rho_0\nu r}{(8/3)\pi\rho_0 r^3} = 6\cdot\dfrac{3}{8}\cdot\dfrac{\nu}{r^2} = \dfrac{9\nu}{4r^2}$, so

$$\ddot{x} + \frac{9\nu}{4r^2}\dot{x} = \frac{g}{2} \quad \Longleftrightarrow \quad \boxed{4r^2\ddot{x} + 9\nu\dot{x} = 2r^2 g}$$

- One unknown, $x(t)$, and one linear equation with constant coefficients.

3.4 Integration and Answers

(a) Terminal velocity

> **Pedagogical Justification:** the terminal velocity is reached when the acceleration vanishes ($\ddot{x} = 0$), i.e. when the drag balances the buoyancy-corrected weight:

$$9\nu v_t = 2r^2 g \implies \boxed{v_t = \frac{2r^2 g}{9\nu}}$$

(b) Position as a function of time

- **Velocity.** Let $\gamma = \dfrac{9\nu}{4r^2}$ ($[\gamma] = \mathrm{s}^{-1}$). Then $\dot{v} + \gamma v = \dfrac{g}{2} = \gamma v_t$, i.e. $\dfrac{dv}{v - v_t} = -\gamma\, dt$. Integrating from $v(0) = 0$ with the substitution $w = v' - v_t$, $dw = dv'$, whose antiderivative is $\int dw/w = \ln|w|$ (Barrow):

$$\int_0^v \frac{dv'}{v' - v_t} = -\gamma \int_0^t dt' \implies \left[\ln|v' - v_t|\right]_0^v = \ln|v - v_t| - \ln|0 - v_t| = \ln\frac{v_t - v}{v_t} = -\gamma t$$

(the absolute value is removed because $v < v_t$: the net force $m\gamma(v_t - v)$ is positive while $v < v_t$, so v increases from 0, and it cannot reach v_t in finite time because $v \equiv v_t$ is itself a solution and the solutions of this linear equation are unique). Exponentiating, $1 - v/v_t = e^{-\gamma t}$:

$$v(t) = v_t\left(1 - e^{-\gamma t}\right) > 0 \quad \text{for } t > 0$$

so $\dot{x} > 0$ at all times and the drag indeed points upward, as assumed in 3.2.

- **Position.** Integrating the velocity with $x(0) = 0$:

With $u = -\gamma t'$, $du = -\gamma\, dt'$, the antiderivative is $\int e^{-\gamma t'}\, dt' = -\frac{1}{\gamma}\int e^{u}\, du = -\frac{1}{\gamma}e^{-\gamma t'}$, so

$$x(t) = \int_0^t v_t\left(1 - e^{-\gamma t'}\right) dt' = v_t\left[t' + \frac{1}{\gamma}e^{-\gamma t'}\right]_0^t = v_t\left(t + \frac{1}{\gamma}e^{-\gamma t} - \frac{1}{\gamma}\right) = \frac{v_t}{\gamma}\left[\gamma t - 1 + e^{-\gamma t}\right]$$

- Substituting $\frac{v_t}{\gamma} = \frac{2r^2 g}{9\nu}\cdot\frac{4r^2}{9\nu} = \frac{8r^4 g}{81\nu^2}$ and $\gamma = \frac{9\nu}{4r^2}$, and writing $\frac{8r^4 g}{81\nu^2} = \frac{2r^2 g}{81\nu^2}\cdot 4r^2$ so that $4r^2$ multiplies the bracket (with $4r^2\cdot\gamma t = 4r^2\cdot\frac{9\nu}{4r^2}t = 9\nu t$):

$$\boxed{x(t) = \frac{2r^2 g}{81\nu^2}\left[-4r^2 + 4r^2\exp\left(-\frac{9\nu}{4r^2}t\right) + 9\nu t\right]}$$

Interpretation & Checks

- **Short times ($t \ll \gamma^{-1}$):** with $e^{-\gamma t} \approx 1 - \gamma t + \frac{1}{2}\gamma^2 t^2$, $x \approx \frac{v_t}{\gamma}\cdot\frac{1}{2}\gamma^2 t^2 = \frac{1}{2}(\gamma v_t)t^2 = \frac{1}{4}gt^2$: free fall under the effective gravity $g\,(1 - \rho_0/\rho_s) = g/2$.
- **Long times ($t \gg \gamma^{-1}$):** $x \approx v_t t - v_t/\gamma$, uniform descent at v_t with an offset v_t/γ.
- **Dimensional check:** $[v_t] = \dfrac{\mathrm{m}^2\cdot\mathrm{m\,s}^{-2}}{\mathrm{m}^2\mathrm{s}^{-1}} = \mathrm{m/s}$ and $[\gamma] = \mathrm{m}^2\mathrm{s}^{-1}/\mathrm{m}^2 = \mathrm{s}^{-1}$.
- **Check against the official key:** the key of `Problems.pdf` gives the equation of motion $4r^2\ddot{x} + 9\nu\dot{x} = 2r^2 g$ (with x positive downward, initial conditions $x(0) = \dot{x}(0) = 0$) and $x = \frac{2r^2 g}{81\nu^2}\left[-4r^2 + 4r^2\exp\left(-\frac{9\nu}{4r^2}t\right) + 9\nu t\right]$. Both are identical to the results above. Numerical integration (DOP853, $g = r = \nu = 1$) gives $x(2) = 0.3467762$, equal to the closed form.

Source: Problems.pdf